



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

SURVEYING TECHNICAL REPORT

LSVTR302

Produce surveying technical report

Competence

RQF Level: 3

Learning Hours



Credits: 4

Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land surveying

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Issue Date: May 2022

Purpose statement	This module titled leveling Surveying technical report was developed for students of RQF Level 3 land surveying for the purpose of equipping them with the skills, knowledge and attitude required to produce a quality and accurate surveying technical report which describes the surveying work done.				
Learning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language; Linear distance measurement; Construction drawings interpretation; Building setting out; Angular measurements.				
Delivery modality	Training delivery		100%	Assessment	
	Theoretical content		30%		
	Practical work:		70%		50%
	• Group project and presentation	20%			
	• Individual project /Work	50%			
			Summative Assessment		50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe surveying technical reports	1.1. Surveying technical reports are properly identified according to their categories.
	1.2. Parts of surveying technical report are properly identified according to the report format.
	1.3. Parts of surveying technical report are properly described according to the report format.

2. Identify types of Surveying projects, instrument and data	2.1. Surveying Project types are properly identified according to the type of work.
	2.2. Instruments are properly identified according to their types.
	2.3. Surveying data are properly identified according to their types.
	2.4. Site information is properly recorded according to the workplace.
3. Prepare and Present surveying technical report	3.1. Types of surveying report are properly identified according to the work.
	3.2. Surveying report is properly produced according to the work done.
	3.3. Surveying technical report is properly presented according to the method of presentation

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Distinguish surveying technical report 2. Identify types of Surveying projects, instrument and data 3. Prepare and Present surveying technical report
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Learning outcome 1: Describe surveying technical report	Learning hours: 9
Indicative content	
* Categories of surveying technical reports • Field reports • Office reports * Identification of parts of surveying technical report • Title page, abstract, table of contents, list figures, list tables, abbreviation,	

introduction, body, conclusion, references, appendices, etc)	
* Description of parts of surveying technical report Title page, abstract, table of contents, list figures, list tables, abbreviation, introduction, body, conclusion, references, appendices, etc)	
Resources required for the learning outcome	
Equipment	Computer, projector, scanner, printer
Materials	Pieces of paper, marker pens, ink, flipchart.
Tools	Flash disk, USB cables, pens, pencils.
Facilitation techniques	- Demonstration and simulation - Individual and group work - Individualized - Trainer guided - Group discussion
Formative assessment methods	- Written assessment - Oral assessment - Performance assessment - Product based assessment

Learning outcome 2: Identify types of Surveying projects, instrument and data	Learning hours: 11
Indicative content	
* Surveying Project types * Identification of surveying instruments - Level, theodolite, total station, GPS/GNSS, laser meter, etc. * Identification of surveying data - Coordinates, angles, distances, volumes, etc. * Identification of site information - Weather condition, location, challenges, etc.	
Resources required for the indicative content	

Equipment	computer, projector , level, theodolite, total station, GPS/GNSS, laser meter
Materials	Pieces of paper, pens, markers, ink pens, flipchart, internet.
Tools	Flash disk, USB cables, etc.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 3: Prepare and Present surveying technical report	Learning hours: 20
Indicative content	
* Identification of types of surveying report ○ Cadastral surveying report, ○ Levelling surveying report, ○ Topographic surveying report, ○ Mining surveying report * Production of surveying technical report ○ Report structure, drafting, proofreading * Presentation of surveying technical report ○ visual aids: Power point presentation, flip chart or white board, printed documents etc.	
Resources required for the indicative content	

Equipment	Computer, projector , level, theodolite, total station, GPS/GNSS, laser meter
Materials	Pieces of paper, pens, marker pens, ink pens, flipchart, internet, etc
Tools	Flash disk, USB cables, etc.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment • Project-based assessment

Integrated/Summative assessment

Integrated situation

The government of Rwanda has initiated the policy of land management by residing inhabitants into settlements in place of scattered homes. This provides easy access to public infrastructures among others. In line with the government policy, Gakenke district offered a tender for producing the physical plan of Rugaragara model village in Rusagara cell and Rhino Construction Ltd has become the successful bidder. Relevant data have been collected and cadastral maps completed. As the company surveyor, your task is to prepare a technical report and present it to the district authority prior to the approval of the physical plan. Here are important data to be considered in your report:

- Identification of landowners
- Instruments used

- Number of parcels after subdivision
- Affected parcels
- Identification of the parcels neighboring the study area

Resources

Equipment	Computer, projector , level, theodolite, total station, GPS/GNSS, laser meter
Materials	Pieces of papers, pens, marker pens, ink pens, flipchart, internet, et.c
Tools	Flash disk, USB cables, etc
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion Etc...
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment Etc...

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Identify surveying technical reports 30%	1.2 Parts of surveying technical report are properly described	1. Title page is done.			3
		2. Abstract is done.			2
		3. Table of content is done.			2
		4. List of figures is well done.			2

	according to the report format	5. List of tables is well done.			3
		6. Abbreviation is well done.			3
		7. Introduction is well done.			5
		8. Body is well done.			10
		9. Conclusion is well done.			5
		10. Reference is well done.			5
		11. Appendices are well done.			5
2. Identify types of instrument and data 25%	2.1 Surveying data are properly identified according to the instruments used	1. Instrument is well specified			3
		2 Types of data are well specified.			4
		3. The data is well arranged			3
	2.2. Site information are properly identified according to the workplace	1. Weather the condition is well specified.			5
		2 Location is well specified.			5
		3. Challenges are well mentioned			
	3.2. Surveying report are properly produced	1 Report structure is well organized.			10

	according to the work	2 Drafting is well organized.			5
		3 Proofreading is well done.			5
	3. 3 Surveying technical report are properly presented according to the work	1. Graphics features are well organized.			5
		2. Drawings are well organized.			5
		3. PowerPoint presentation is well organized.			5
	Total marks		100		
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

Genovese I. (2005), Definition of surveying Terms, ACSM, 314pp, ISBN 0-9765991-0-4.